DC Charging (CCS2)-V

Introduction

The rapid adoption of electric vehicles (EVs) necessitates robust and intelligent charging infrastructure. At the heart of advanced DC charging systems, such as CCS2, lies a sophisticated communication framework that ensures safe, efficient, and reliable power transfer. This lecture delves into the intricate high-level communication (HLC) protocols that govern the interaction between an Electric Vehicle (EV) and an Electric Vehicle Supply Equipment (EVSE), building upon the foundational understanding of the 7-layer Open Systems Interconnection (OSI) model.

Understanding these communication layers is paramount for anyone involved in the design, implementation, or maintenance of EV charging systems. From the initial physical connection to the final power delivery and session termination, every step is orchestrated through a series of precisely defined messages and protocols. This lecture will dissect how these layers work in concert, enabling seamless data exchange, error control, and secure transactions, ultimately facilitating the widespread and dependable operation of DC fast charging for the next generation of electric mobility.

Core Concepts

Term	Definition	Significance
OSI Model	A conceptual framework that standardizes the functions of a telecommunication or computing system into seven distinct layers.	Provides a structured approach to understanding and designing complex communication protocols, ensuring interoperability and reliability in CCS2.
HLC	High-Level Communication refers to the advanced digital data exchange between an EV and EVSE, typically over the Control Pilot (CP) and Protective Earth (PE) lines.	Essential for managing complex charging parameters, authentication, and safety checks that cannot be handled by simple PWM signaling.

Term	Definition	Significance
SLAC Procedure	Signal Level Attenuation Characterization, a process used to establish a logical connection between PLC modems in the EV and EVSE.	Ensures that the correct PLC modems are logically connected and ready for data exchange, crucial for multi-modem charging stations.
SECC Discovery Protocol (SDP)	A UDP-based protocol used by the EV to locate and obtain the IP address of the Supply Equipment Communication Controller (SECC).	Allows the EV to find the central communication controller of the EVSE, which might be located remotely, enabling communication setup.
V2GTP	Vehicle-to-Grid Transport Protocol, a session layer protocol specifically designed for EV-EVSE communication.	Manages the establishment, maintenance, and termination of communication sessions, including protocol versioning and payload information.
EXI Format	Efficient XML Interchange, a compact binary representation of XML data used for data encoding and decoding in the presentation layer.	Optimizes data transfer by reducing message size, making communication faster and more efficient between EV and EVSE.
TCP/UDP/ TLS	Transmission Control Protocol (reliable), User Datagram Protocol (fast), and Transport Layer Security (secure) are transport layer protocols.	Provide the necessary mechanisms for reliable, fast, and secure data transmission between the EV and EVSE nodes.
IPv6	Internet Protocol version 6, used for logical addressing and routing in the network layer.	Assigns unique IP addresses to all communicating nodes (EV, EVSE, controllers), enabling data routing across the network.

Term	Definition	Significance
EV as Master	In CCS2 protocol messaging, the Electric Vehicle (EV) always initiates requests, acting as the master, while the EVSE responds as the slave.	Simplifies the communication flow and ensures a clear hierarchy in message exchange, preventing conflicts.
`EVSE processing`	A status field within response messages indicating whether the EVSE is `ongoing` with a process or has `finished` it.	Provides real-time feedback to the EV about the EVSE's operational status during critical phases like parameter discovery or cable checks.

Detailed Analysis

1. The 7-Layer OSI Model in CCS2

The CCS2 DC charging system leverages the 7-layer OSI model to structure its high-level communication, ensuring robust, reliable, and secure data exchange. This model is primarily defined by the ISO 15118 standard, with different parts addressing specific aspects: ISO 15118-1 for general information and use-case definition, ISO 15118-2 for technical protocol descriptions and requirements, and ISO 15118-3 for physical layer requirements.

- **Physical Layer (OSI Layer 1):**
- **Initial Condition:** The charger plug is physically connected to the EV inlet, establishing connections for the Control Pilot (CP) and Protective Earth (PE) lines. A Pulse Width Modulation (PWM) signal with a duty ratio of 3-7% is initiated over the CP line, signaling the vehicle's entry into state B2 and readiness for HLC.
- **Function:** This layer is responsible for stabilizing the physical link on the CP and PE wires. It utilizes HomePlug Green PHY compatible PLC modems on both the EV and EVSE sides.
- **Outcome:** The PLC modules are ready for communication with an established frequency band of 2 MHz to 28 MHz. This layer defines physical properties like voltage levels and frequencies, ensuring the foundational physical connection for data transfer.
- **Data Link Layer (OSI Layer 2):**

- **Initial Condition:** Successful establishment of the physical layer, meaning PLC modems are physically connected.
- **Function:** This layer configures the PLC nodes and performs the Signal Level Attenuation Characterization (SLAC) procedure. SLAC ensures that the correct PLC modems of the EV and EVSE establish a logical connection.
- **Outcome:** A stable communication link is established, ensuring error-free transfer of data frames between nodes over the physical media.
- **Network Layer (OSI Layer 3):**
- **Initial Condition:** Successful establishment of the data link layer, with PLC modems logically connected.
- **Function:** Implements the IPv6 Internet Protocol Standard to define IP addresses for all communicating nodes (EV, EVSE, SECC, EVCC). It uses the Stateless Address Auto Configuration (SLAAC) procedure for IP address assignment and ICMPv6 (Internet Control Message Protocol version 6) for sending error messages.
- **Outcome:** All nodes obtain valid and unique IP addresses, and the layer controls the routing of connections, determining the optimal path for data.
- **Transport Layer (OSI Layer 4):**
- **Initial Condition:** All network layer protocols are successfully stabilized, and all nodes have retrieved their IP addresses.
- **Function:** This layer ensures reliable, fast, and secure communication. It incorporates Transmission Control Protocol (TCP) for reliability, User Datagram Protocol (UDP) for speed, and Transport Layer Security (TLS) for security (optional but often incorporated).
- **Outcome:** A reliable, fast, and safe data connection is established between all communicating nodes (PLC modems, SECC, EVCC).
- **Session Layer (OSI Layer 5):**
- **Initial Condition:** Successful establishment of the transport layer protocol.
- **Function:** Implements the Vehicle-to-Grid Transport Protocol (V2GTP). It establishes, manages, and terminates connections between nodes using IP addresses and port numbers. It also implements a session layer header containing basic information like protocol version, payload type, and payload size.
- **Outcome:** A stable charging session connection is established between the EV and EVSE, facilitating data exchange.
- **Presentation Layer (OSI Layer 6):**

- **Initial Condition:** Successful establishment of the session layer, with both EV and EVSE using the Efficient XML Interchange (EXI) encoding format.
- **Function:** Handles the coding and decoding of data from the application to the network layer format. The V2G EXI protocol includes two message sets: V2G application layer protocol handshake messages and V2G application layer messages. It also implements a message header containing session ID, notification, and signature, along with the message body.
- **Outcome:** Ensures the validity and compatibility of exchanged messages, acting as a translator to enable correct data exchange between EV and EVSE.
- **Application Layer (OSI Layer 7):**
- **Initial Condition:** Successful establishment of all presentation layer protocols.
- **Function:** Manages all messages exchanged between the EV and EVSE, overseeing the entire charging process (identification, pre-charge, charge, security checks). It uses application protocol messages to select the charging standard (e.g., DIN SPEC 70121 or ISO 15118). The EV sends a `supportedAppProtocolReq` listing supported standards, and the EVSE responds with a `supportedAppProtocolRes` selecting one based on a scheme ID.
- **Outcome:** Represents the messages and required communication protocols, ensuring the initialization and configuration of the charge process.

2. SECC Discovery Protocol

After the physical connection is made and logical connection (SLAC) is established, the EV needs to locate the Supply Equipment Communication Controller (SECC). The SECC might be physically located within the EVSE or in a remote data center.

- **Purpose:** To obtain the IP address of the SECC.
- **Mechanism:** The EV uses the SECC Discovery Protocol (SDP), which is a UDP-based protocol, ensuring fast communication.
- The EV broadcasts an `SDP Request` message, which includes a session ID.
- The SECC, upon receiving the request, responds with an `SDP Response` message. This response contains the SECC's IP address, the session ID, and the SECC's TCP port, which will be used for subsequent communication.
- **Significance:** This step is crucial for the EV to establish direct communication with the EVSE's central controller, regardless of its physical location.

3. High-Level Communication (HLC) Sequence

The overall HLC process in CCS2 follows a structured timeline, moving from initial physical signaling to complex data transfer:

- **Control Pilot Signaling:** Begins with IEC 61851-1 compliant 1 kHz PWM signaling, transitioning to a 3-7% duty cycle to indicate HLC readiness.
- **Network Setup:** Involves the physical layer (HomePlug GreenPHY compatible PLC modems) and data link layer (SLAC operation) to establish logical connections. The network layer then uses SLAAC to assign IPv6 addresses to EV and EVSE modems, and SDP to assign an IP address to the SECC.
- **Transport Setup:** Utilizes TCP for reliable and TLS for secure data transmission.
- **Security Setup:** Further defines security protocols, often involving TLS.
- **Session Setup:** Activates the V2G Transport Protocol (V2GTP) and prepares for data formatting using XML or EXI.
- **Data Transfer:** The actual exchange of messages between EV and EVSE controllers via their respective modems.

4. CCS2 Protocol Messaging (ISO 15118/DIN SPEC 70121)

The communication between the EV and EVSE is based on a simple request-response protocol, where the EV always acts as the master, initiating requests, and the EVSE acts as the slave, providing responses. If a critical error occurs or an incorrect request is made, the EVSE will respond with a `FAILEDSequenceError` code, and the communication session will be immediately closed.

- **Communication Setup Sequence:**
- **Application Protocol Selection:** After PLC modems are logically connected, the EV sends a `supportedAppProtocolReq` message, listing all charging standards it supports. The EVSE selects one (e.g., DIN SPEC 70121 or ISO 15118) and responds with a `supportedAppProtocolRes` containing the selected scheme ID.
- **Session Setup:** The EV initiates a new session by sending a `SessionSetupReq` with `sessionID = 0`. The EVSE responds with a new, unique `sessionID`, which is then used for the entire communication. If the EV sends an invalid `sessionID` (e.g., during a session restart), the EVSE responds with a `response code = FAILED`. If a valid old `sessionID` is sent during a restart, the EVSE responds with `response code = OK_OldSession`.
- **Identification, Authentication, and Authorization Sequence:**

- **Service Discovery:** The EV sends a `ServiceDiscoveryReq` to inquire about available services (e.g., EV charging, payment, certificate installation). The EVSE responds with a list of services.
- **Service Details:** The EV can then request specific details about a service using `ServiceDetailReq`, to which the EVSE responds with `ServiceDetailRes` containing all relevant parameters.
- **Payment Service Selection:** The EV requests the type of payment (e.g., contract-based, external) via `PaymentServiceSelectionReq` (ISO 15118) or `ServicePaymentSelection` (DIN SPEC 70121), and lists services to be used for payment.
- **Authorization:** The EV sends an `AuthorizationReq` (ISO 15118) or `ContractAuthentication` (DIN SPEC 70121) message. The EVSE evaluates the authorization (e.g., via RFID card) and responds, indicating whether the session is authorized. The EV waits for this authorization.
- **Target Setting and Charge Scheduling:**
- **Charge Parameter Discovery:** The EV sends `ChargeParameterDiscoveryReq` multiple times to exchange charging parameters like maximum voltage, current, and power. The EVSE processes these and responds. The `EVSE processing` field in the response indicates `ongoing` if parameters are still being determined, or `finished` once they are set. This transitions the charging state to C.
- **Charging Loop / Re-scheduling:**
- **Cable Check:** The EV continuously sends `CableCheckReq` to monitor insulation resistance. The EVSE responds with `CableCheckRes`, with `EVSE processing` indicating `ongoing` or `finished` for the insulation check.
- **Pre-Charge:** The EV sends `PreChargeReq` until it measures a voltage on the EVSE side that matches its battery voltage within a 20V tolerance. The EVSE responds with its current output voltage. This prevents large inrush currents.
- **Power Delivery:** After successful pre-charging, the EV sends `PowerDeliveryReq`. Upon receiving `PowerDeliveryRes`, the EV closes its contactor switch and proceeds with charging.
- **Current Demand:** During active charging, the EV continuously sends `CurrentDemandReq`, specifying desired charging voltage and current based on its battery's SOC. The EVSE processes these values, adjusts its power source, and responds with the actual voltage and current. The EVSE can also trigger a stop of charging by sending a notification within the `CurrentDemandRes`.

- **End of Charging:**
- **Power Delivery (End):** When the battery reaches the desired SOC, the EV stops the charging process by sending a `PowerDelivery` message. This changes the EV's state to B.
- **Welding Detection (Optional):** The EV sends a `WeldingDetectionReq` to check if the DC contactors have opened. The EVSE responds with `WeldingDetectionRes` containing the measured voltage of the DC lines. If the voltage is not zero, it indicates a potential welding issue, and an error message is initiated.
- **Session Stop:** The EV initiates a `SessionStopReq`, which the EVSE responds to, terminating the communication session. The charger plug can then be safely removed.

Key Takeaways

- [ ] The 7-layer OSI model provides a structured framework for CCS2 high-level communication, ensuring reliability and interoperability.
- [ ] High-level communication (HLC) is essential for complex DC charging, utilizing PLC over CP and PE lines, operating between 2-28 MHz.
- [ ] The EV acts as the master, initiating all requests, while the EVSE responds as the slave in the CCS2 protocol messaging.
- [ ] The SLAC procedure is critical for establishing a logical connection between the correct PLC modems of the EV and EVSE.
- [ ] The SECC Discovery Protocol (SDP) allows the EV to locate and obtain the IP address of the EVSE's communication controller.
- [ ] Transport layer protocols (TCP, UDP, TLS) ensure reliable, fast, and secure data transmission during charging.
- [ ] The charging process involves a precise sequence of message exchanges, including session setup, service discovery, authorization, parameter negotiation, and continuous current demand.
- [ ] Safety checks like Cable Check (insulation resistance) and Pre-Charge (voltage matching within 20V) are integral to the charging sequence.
- [ ] The `EVSE processing` field provides crucial status updates (ongoing/finished) from the EVSE during various stages of communication.
- [ ] Critical errors lead to immediate session termination, highlighting the importance of following the defined communication sequence.

Conclusion

The detailed exploration of the CCS2 DC charging system's high-level communication protocols underscores the complexity and sophistication required to manage modern EV charging. From the foundational OSI model layers to the specific request-response messages, every element is meticulously designed to ensure a secure, efficient, and user-friendly charging experience. This intricate dance of data, orchestrated over the control pilot and protective earth lines, is what transforms a simple electrical connection into an intelligent energy transfer system.

As electric mobility continues to evolve, these communication standards will serve as the bedrock for future innovations, including vehicle-to-grid (V2G) capabilities and advanced energy management. Understanding these protocols is not just about comprehending how current systems work, but also about preparing for the next generation of smart charging solutions that will further integrate EVs into the broader energy ecosystem. The principles discussed here will be instrumental in developing and deploying the charging infrastructure of tomorrow.